

Minimum Colors for Non-Perfect Coloring of the Eleven Euclidean Uniform Tilings

Keziah Nicole C. Lingahan, Michaela Faye J. Palines, Sieg Gabriel G. Sola

ABSTRACT

This study investigates the relationship between color symmetry and geometric structure in the eleven Euclidean uniform tilings. A coloring is considered perfect if the tiling's symmetries preserve the color pattern through transformations. The primary objective is to determine the minimum number of colors required to guarantee a non-perfect coloring where symmetry is necessarily broken for each tiling. The methodology utilizes group theory, specifically analyzing transitive tile orbits and stabilizer subgroups. Orbits are classified into Type I, sharing the symmetry of regular tilings, and Type II, requiring a derived index formula based on point group and translation subgroup indices. By generating the complete set of perfect coloring numbers for each tiling, the minimum non perfect coloring number is identified as the smallest integer not present in that set. Results show that regular tilings require minimal colors (2 or 3) to force non-perfection. However, tilings with multiple orbits require significantly more. The 4.8.8 tiling requires 7 colors, while the 3.6.3.6 and 3.12.12 tilings require 23 colors to ensure a non-perfect coloring. These findings quantify the threshold at which color symmetry is lost and provide a structured framework for analyzing non-symmetry in mathematical patterns.

Keywords: *color symmetry, Euclidean uniform tilings, orbits, non-perfect coloring, stabilizer subgroups*

1. Introduction

According to Grünbaum and Shephard (1987), a plane tiling or tessellation is a collection of sets that covers the plane completely without gaps or overlaps. Tiling theory studies how such arrangements of tiles are formed and examines their mathematical properties, particularly symmetry. One important class is uniform tilings, where the same configuration of regular polygons occurs around every vertex throughout the plane (Grünbaum and Shephard 1987). Because of their high degree of regularity, uniform tilings provide a natural setting for investigating symmetry in geometric patterns.

Coloring is a fundamental tool for analyzing symmetry in tilings. Grünbaum and Shephard's work initiated the systematic study of perfect colorings, establishing connections between the number of colors used and the structure of the symmetry group. A coloring is said to be perfect when every symmetry of the uncolored tiling corresponds to a permutation of the color set, so that the overall colored pattern is preserved under the action of the plane crystallographic group (Kaplan 2022). This means that transformations such as rotations or reflections may interchange colors, provided the interchange is consistent across the tiling.

There are eleven distinct Euclidean uniform tilings, each associated with a plane crystallographic group or symmetry group that captures its symmetries. For these tilings, perfect colorings may exist in multiple forms, depending on the number of colors and the induced permutations. However, perfect coloring is not always possible once the number of colors increases beyond certain limits. This creates an open question about when symmetry can no longer be maintained through coloring.

This study addresses this gap by determining, for each of the eleven Euclidean uniform tilings, the minimum number of colors η required such that no perfect coloring can be achieved. By deriving the perfect coloring numbers of each tiling, the research identifies the threshold at which non-perfect coloring becomes unavoidable. This contributes to tiling theory by clarifying how color constraints interact with symmetry groups and by providing a structured group-theoretic perspective through the use of orbits.

The scope of the study is limited to the eleven Euclidean uniform tilings and their corresponding plane crystallographic groups. The focus is on the number of colors required to guarantee non-perfect colorings, rather than on classifying the full color-permutation groups associated with each tiling.

2. Materials & Methods

This study employs theoretical and computational group-theoretic methods to determine, for each of the eleven Euclidean uniform tilings, the minimum number of colors η that guarantees a non-perfect coloring. The general procedure consists of first generating the complete set of all possible perfect coloring numbers K for a given tiling, and then identifying the smallest integer not contained in the set.

The central concept used throughout is the perfect coloring number k . A coloring of a periodic tiling is defined as perfect when the symmetry group (plane crystallographic group) $\mathcal{G}$ of the uncolored tiling induces a valid permutation of the color classes. In this sense, every symmetry of the tiling corresponds to a consistent relabeling of colors, ensuring that the overall colored pattern remains invariant under the action of $\mathcal{G}$.

2.1. Perfect Coloring Numbers for Regular Tilings

The methodological foundation of the study relies on the established classification results of Grünbaum and Shephard (1977) for the three regular Euclidean tilings. Their sets of possible perfect coloring numbers are adopted as axiomatic starting points. In addition, they proved that every transitive tiling admits perfect k -coloring for infinitely many integers k . The relevant classification provide the baseline for extending the analysis to all uniform tilings and is given by:

$$K(3.3.3.3.3.3) = 2n^2, 6n^2, (3n - 2)^2, (3n - 1)^2; n \geq 1$$

$$K(4.4.4.4) = n^2, 2n^2; n \geq 1$$

$$K(6.6.6) = n^2, 3n^2; n \geq 1$$

2.2. Orbit Decomposition of Uniform Tilings

Uniform tilings that contain more than one transitive orbit of tiles were analyzed by decomposing the tiling into independent tile orbits under the action of the symmetry group $\mathcal{G}$. Each orbit is treated separately, since perfect coloring must be compatible across all orbits of the tiling.

For a tile $t \in \mathcal{T}$, its orbit is defined as:

$$\mathcal{O}(t) = \{g(t) | g \in \mathcal{G}\}$$

2.3. Classification of Orbit Types

Each tile orbit was classified into one of two categories: Type I and Type II. Type I orbits are those that share both the same symmetry group as a regular tiling and the same tile stabilizer. In this case, the perfect coloring numbers coincide directly with the corresponding regular tiling classification. Otherwise, the orbits is considered as Type II. For such orbits, the admissible perfect coloring numbers were derived using a general formula which is the product of the index of the point group, the index of the translation subgroup, and an appropriate scaling function. This provides the necessary extension beyond the regular tiling cases.

2.4. Perfect Coloring Number Formulas for Type I and Type II Orbits

For each tile orbit identified in the decomposition of a uniform tiling, the set of admissible perfect coloring numbers is determined according to its orbit type. Two cases arise, depending on the relationship between the orbit stabilizer and the parent symmetry group.

For Type I orbits, the admissible perfect coloring numbers are inherited directly from regular tilings. This follows from the fact that such orbits share both the same parent symmetry group $\mathcal{G}$ and the same tile stabilizer subgroup $\mathcal{S}$ as a corresponding regular tiling.

Proposition 2.4.1. Let $\mathcal{O}_1$ and $\mathcal{O}_2$ be two transitive orbits of tiles in the Euclidean plane sharing the same parent symmetry group $\mathcal{G}$. If the stabilizer subgroup $\mathcal{S}_1$ of a tile in $\mathcal{O}_1$ is the same as the stabilizer subgroup $\mathcal{S}_2$ of a tile in $\mathcal{O}_2$, then the set of possible perfect coloring numbers for $\mathcal{O}_1$ is identical to that of $\mathcal{O}_2$.

A perfect coloring of a transitive tile orbit $\mathcal{O}$ is defined by a subgroup $\mathcal{J} \leq \mathcal{G}$ such that the index $k = [\mathcal{G} : \mathcal{J}]$ is finite, subject to the condition:

$$\mathcal{S} \leq \mathcal{J} \leq \mathcal{G},$$

where $\mathcal{S} = \text{Stab}_{\mathcal{G}}(t)$ is the stabilizer of a representative tile $t \in \mathcal{O}$. The admissible perfect coloring numbers are therefore:

$$K = \{ [\mathcal{G} : \mathcal{J}] \mid \mathcal{S} \leq \mathcal{J} \leq \mathcal{G} \}$$

When two orbits share the same symmetry group and stabilizer, the lattices of admissible subgroups $\mathcal{J}$ coincide, implying identical sets of indices. Consequently, for Type I orbits, the perfect coloring numbers coincide exactly with those of the associate regular tiling.

Proposition 2.4.2. For an orbit of tiles with parent symmetry group $\mathcal{G}$ and valid coloring subgroup $\mathcal{J}$, the perfect coloring numbers or index $k = [\mathcal{G} : \mathcal{J}]$ is determined by the product of the point group index, translation subgroup index, and a scaling function:

$$k = I_p \cdot (I_t \cdot f(n))$$

where $I_p = [P_{\mathcal{G}} : P_{\mathcal{J}}]$ and the product $(I_t \cdot f(n))$ corresponds to the lattice index $[T_{\mathcal{G}} : T_{\mathcal{J}}]$.

The index decomposition reflects the geometric structure of symmetry groups. The factor I_p counts the number of distinct orientations required when coloring subgroup has reduced point symmetry, while the translational factor $I_t \cdot f(n)$ measures the number of fundamental translation cells of $\mathcal{G}$ contained within a fundamental cell of $\mathcal{J}$. The term I_t accounts for minimal lattice alignment, and $f(n)$ represents integer scaling of lattice vectors.

2.5. Construction of the Complete Perfect Coloring Set

After determining the admissible perfect coloring numbers for each individual tile orbit using the formulas for Type I and Type II cases, the full set of perfect coloring numbers for the entire uniform tiling is constructed by combining the orbit-level results.

Let $\mathcal{O}_1, \mathcal{O}_2, \dots, \mathcal{O}_m$ denote the distinct tile orbits of a tiling $\mathcal{T}$ under the action of its symmetry group $\mathcal{G}$. For each orbit $\mathcal{O}_i$, let $\mathcal{C}_i$ represent the set of all valid perfect coloring numbers associated with that orbit.

A perfect coloring of the full tiling must be compatible across all orbits simultaneously. Therefore, the set of admissible perfect coloring numbers for the complete tiling, denoted by K , is obtained through the Minkowski sum of the orbit-level sets:

$$K = \{k_1 + \dots + k_m \mid k_i \in \mathcal{C}_i\}.$$

This construction represents every possible combination of valid orbit colorings, and thus characterizes all integers k for which a perfect coloring of the entire tiling is theoretically possible.

Once the set K is established, the minimum number of colors required to guarantee a non-perfect coloring is defined as the smallest integer not in the set K :

$$\eta = \min(k \notin K \mid k \geq 2)$$

In particular, colorings using a number of colors less than or equal to the number of independent orbits are considered trivial perfect colorings. The value of η therefore marks the first nontrivial threshold beyond which perfect symmetry-preserving colorings cannot occur.

This procedure was applied systematically to each of the eleven Euclidean uniform tilings in order to determine the corresponding minimum non-perfect coloring number.

3. Result and Discussion

3.1. Case 1: Tilings with one orbit (regular tilings)

The perfect coloring numbers for these fundamental structures are taken as axiomatic based on the theorems of Grünbaum and Shephard. The following tilings are regular tilings and each have one orbit of tiles only.

3.1.1. The regular tiling 3.3.3.3.3 ($\mathcal{G} \cong p6m, \mathcal{S} \cong D_3$)

Perfect coloring numbers: $k = 2n^2, 6n^2, (3n - 2)^2, (3n - 1)^2; n \geq 1$

Admissible values:

$$k = 2n^2: k = \{2, 8, 18, 32, 50, \dots\}$$

$$k = 6n^2: k = \{6, 24, 54, 96, \dots\}$$

$$k = (3n - 2)^2: k = \{1, 16, 49, 100, \dots\}$$

$$k = (3n - 1)^2: k = \{4, 25, 64, 121, \dots\}$$

$$K(3.3.3.3.3) = \{1, 2, 4, 6, 8, 16, 18, \dots\}$$

The smallest integer not included in the set, $K(3.3.3.3.3)$, is 3. That is, the minimum number of colors that force a non-perfect coloring for the 3.3.3.3.3 tiling is 3 and $\eta(3.3.3.3.3) = 3$.

3.1.2. The regular tiling 4.4.4.4 ($\mathcal{G} \cong p4m, \mathcal{S} \cong D_4$)

Perfect coloring numbers: $k = n^2, 2n^2; n \geq 1$

Admissible values:

$$k = n^2: k = \{1, 4, 9, 16, 25, \dots\}$$

$$k = 2n^2: k = \{2, 8, 18, 32, 50, \dots\}$$

$$K(4.4.4.4) = \{1, 2, 4, 8, 9, 16, 18, \dots\}$$

The smallest integer not included in the set, $K(4.4.4.4)$, is 3. Hence, $\eta(4.4.4.4) = 3$.

3.1.3. The regular tiling 6.6.6 ($\mathcal{G} \cong p6m, \mathcal{S} \cong D_6$)

Perfect coloring numbers: $k = n^2, 3n^2; n \geq 1$

Admissible values:

$$k = n^2: k = \{1, 4, 9, 16, 25, \dots\}$$

$$k = 3n^2: k = \{3, 12, 27, 48, 75, \dots\}$$

$$K(6.6.6) = \{1, 3, 4, 9, 12, 16, 25, \dots\}$$

The smallest integer not included in the set, $K(6.6.6)$, is 2 and $\eta(6.6.6) = 2$.

3.2. Case 2: Tilings with more than one orbit (non-regular tilings)

The analysis progresses to the uniform tilings characterized by two or more distinct transitive orbits of tiles. In this case, the symmetry group, $\mathcal{G}$, acts transitively on each orbit individually, but no symmetry maps a tile from one orbit to another. Consequently, the coloring constraints for the orbits are mathematically independent. For these tilings, let $\mathcal{O}_1$ be the orbit of polygons with fewest number of sides, $\mathcal{O}_2$ be the next, and so on.

3.2.1. The non-regular tiling 4.8.8 ($\mathcal{G} \cong p4m$)

The 4.8.8 tiling is partitioned into independent orbits based on the action of the symmetry group $\mathcal{G} \cong p4m$. Since symmetry operations are isometries, they cannot map a tile of one geometric shape to another, thus, the square tiles and octagonal tiles necessarily form distinct equivalence classes. Furthermore, because the group $p4m$ acts transitively on the set of all squares and transitively on the set of all octagons, these classes are irreducible. Consequently, the tiling consists of exactly two independent orbits, the orbits of squares ($\mathcal{O}_1$) and the orbit of octagons ($\mathcal{O}_2$).

The orbit of square tiles, $\mathcal{O}_1$, is permutationally isomorphic to the regular 4.4.4.4 tiling. This permutational isomorphism implies that the action of the symmetry group $\mathcal{G}$ on the squares in the 4.8.8 tiling is identical to its action on the tiles of the regular square tiling. Specifically, there exists a one-to-one mapping between the tiles and the symmetry operations such that the color permutations induced by any symmetry $g \in \mathcal{G}$ are preserved. This relationship is rooted in a structural isomorphism, as both tilings share the same parent symmetry group $\mathcal{G} \cong p4m$ and tile stabilizer $\mathcal{S} \cong D_4$, also making the squares' orbit Type I. The structural isomorphism between the two systems is demonstrated in Fig. 1.

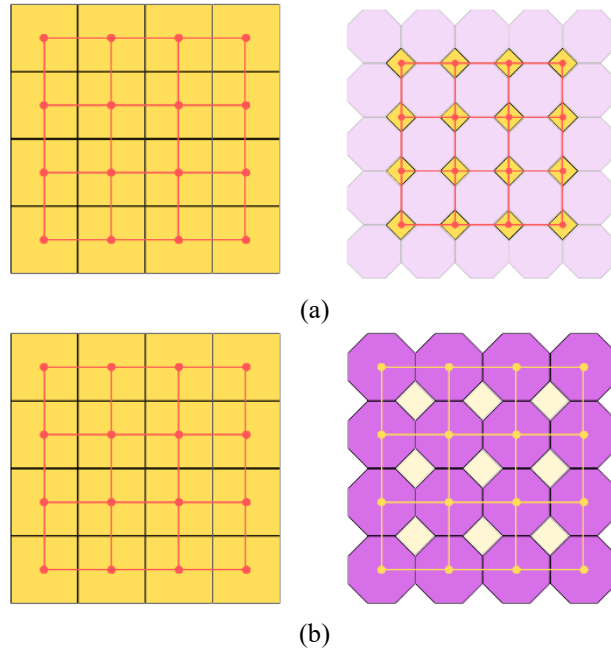


Figure 1. structural isomorphism between the 4.4.4.4 and 4.8.8 tilings

Consequently, the analysis employs the established formula for the 4.4.4.4 tiling, that is:

$$K(4.8.8, \mathcal{O}_1) = K(4.4.4.4) = \{1, 2, 4, 8, 9, 16, 18, \dots\}$$

Similarly, the orbit of octagon tiles, $\mathcal{O}_2$, is permutationally isomorphic to 4.4.4.4. Hence, classified as a Type I orbit.

$$K(4.8.8, \mathcal{O}_2) = K(4.4.4.4) = \{1, 2, 4, 8, 9, 16, 18, \dots\}$$

To identify the minimum number of colors for which no perfect coloring exists, a sequential verification is performed for integers $k \geq 3$ (non-trivial). This k is decomposed into a sum of two components, $k_{\mathcal{O}_1}$ and $k_{\mathcal{O}_2}$.

Table 1. Sequential Verification for k of 4.8.8

Target k	k_{O_1}	k_{O_2}
3	2	1
4	2	2
5	4	1
6	2	4
7	None	None

The integer 7 is the first value that cannot be partitioned into valid components k_{O_1} and k_{O_2} , that is, $\eta(4.8.8) = 7$.

3.2.2. The non-regular tilings 3.6.3.6 and 3.12.12 ($\mathcal{G} \cong p6m$)

The action of the symmetry group $\mathcal{G} \cong p6m$ on the 3.6.3.6 tiling naturally partitions the tiles into independent orbits. Because the symmetry operations are isometries, they preserve the geometric identity of the faces, a triangular tile can never be mapped to a hexagonal one, and vice versa. Furthermore, since $\mathcal{G}$ acts transitively on the set of all triangles and, separately, on the set of all hexagons, these geometric classes are irreducible. Therefore, there are exactly two independent orbits identified, the orbit of triangles and the orbit of hexagons. Similarly, this applies to 3.12.12 as well and the symmetry group $\mathcal{G} \cong p6m$ partitions the tiles into two orbits of triangles and dodecagons.

The two orbits of 3.6.3.6 are both classified as Type I. The orbit of triangle tiles is permutationally isomorphic to 3.3.3.3.3.3 with $\mathcal{G} \cong p6m$ and $\mathcal{S} \cong D_3$ and the orbit of hexagon tiles is permutationally isomorphic to 6.6.6 with $\mathcal{G} \cong p6m$ and $\mathcal{S} \cong D_6$.

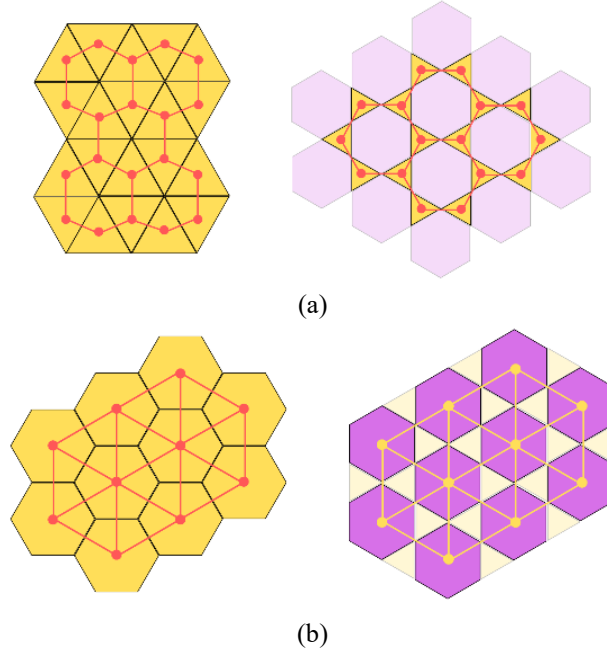


Figure 2. Structural Isomorphism Between the Regular 3.3.3.3.3.3 and 6.6.6 Tilings, and 3.6.3.6

Hence, the perfect coloring numbers of each are as follows:

$$K(3.6.3.6, O_1) = K(3.3.3.3.3.3) = \{1, 2, 4, 6, 8, 16, 18, \dots\}$$

$$K(3.6.3.6, O_2) = K(6.6.6) = \{1, 3, 4, 9, 12, 16, \dots\}$$

Table 2 shows the possible combinations of $k_{\mathcal{O}_1}$ and $k_{\mathcal{O}_2}$ for the perfect coloring numbers of the 3.6.3.6 tiling.

Table 2. Sequential Verification for k of 3.6.3.6

Target k	$k_{\mathcal{O}_1}$	$k_{\mathcal{O}_2}$	Target k	$k_{\mathcal{O}_1}$	$k_{\mathcal{O}_2}$
3	2	1	14	2	12
4	1	3	15	6	9
5	2	3	16	4	12
6	2	4	17	8	9
7	4	3	18	6	12
8	4	4	19	16	3
9	6	3	20	4	16
10	6	4	21	18	3
11	8	3	22	6	16
12	8	4	23	None	None
13	4	9			

The 3.12.12 tiling shares the same symmetry group $\mathcal{G}$ as the 3.6.3.6 tiling, and the tile stabilizers for orbits $\mathcal{O}_1$ and $\mathcal{O}_2$ are identical to those of the 3.6.3.6 tiling. Despite the difference in polygonal boundaries, the centroids of their respective tile orbits occupy identical positions within the symmetry framework (refer to Fig. 3; note that reflections and glide reflections are omitted for clarity).

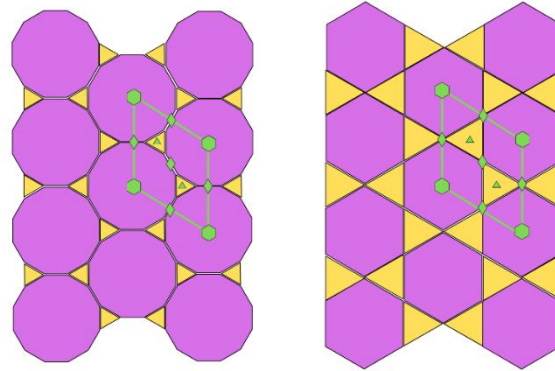


Figure 3. structural isomorphism between 3.12.12 and 3.6.3.6: mapping of rotational centers and translation unit cells

Consequently, these tilings possess a structural isomorphism, implying that the perfect coloring numbers and the underlying coloring configuration of the 3.12.12 tiling are identical to those of the 3.6.3.6 tiling. That is:

$$\eta(3.6.3.6) = \eta(3.12.12) = 23$$

3.2.3. The non-regular tiling 3.3.3.4.4 ($\mathcal{G} \cong cmm$)

The 3.3.3.4.4 tiling is characterized by a parent symmetry group $\mathcal{G} \cong cmm$ and consists of two distinct transitive orbits, an orbit of triangular tiles, denoted as $\mathcal{O}_1$, and an orbit of square tiles, $\mathcal{O}_2$. Both orbits are classified as Type II (structurally unique) because their stabilizers within cmm framework do not map directly to the stabilizers of regular tilings. Consequently, the perfect coloring numbers for both orbits were derived using the general index formula $k = I_p \cdot (I_t \cdot f(n))$. The parent group cmm has a point group index order of 4. For the triangular orbit, the tile stabilizer is D_1 (order 2), while for the square orbit, the tile stabilizer is D_2 (order 4).

The analysis of the triangular orbit considered all subgroups $\mathcal{J}$ of cmm that contain the tile stabilizer D_1 . The compatible subgroups identified were pm , cm , pmm , pmg , and cmm . For the subgroup $\mathcal{J} \cong pm$ (order 2), the point group index was calculated as $I_p = \frac{4}{2} = 2$. The translation index was determined to be 2, as fitting a

primitive rectangular lattice (pm) onto the centered rhombic lattice of $cm\bar{m}$ requires doubling the unit cell area (see Fig. 4). Since primitive rectangular lattice allows for linear scaling, the scaling function is $f(n) = n$, resulting in indices of the form $k_{pm} = 2 \cdot 2 \cdot n = 4n$. Similarly, for $\mathcal{J} \cong cm$, the point group index remained 2 but the lattices are compatible, $I_t = 1$ (Fig. 4). To preserve the centering parity of the lattice, scaling is restricted to odd factors or quadrupled factors, yielding indices $k_{cm} = 2 \cdot 1 \cdot (2n - 1) = 2(2n - 1)$ or $k_{cm} = 2 \cdot 1 \cdot (4n) = 2(4n)$. For the higher-order subgroups pmm , pmg , and $cm\bar{m}$ (all order 4), the point group index becomes $I_p = \frac{4}{4} = 1$. The geometric analysis for these lattices revealed varying translation costs, pmm and pmg required doubling the area ($I_t = 2$) and a linear scaling function leading to indices $2n$, while $cm\bar{m}$ aligns perfectly ($I_t = 1$) and scaling of odd factors or quadrupled factors yielding indices $2(2n - 1)$ and $2(4n)$. Combining these results, the set of perfect coloring numbers for the triangular orbit is the union of all derived sets, summarized as follows:

$$\begin{aligned}
k_{pm}(3.3.3.4.4, \mathcal{O}_1) &= 4n = \{4, 8, 12, 16, 20, \dots\} \\
k_{cm}(3.3.3.4.4, \mathcal{O}_1) &= 2(2n - 1), 2(4n) = \{2, 6, 8, 10, 14, \dots\} \\
k_{pmm}(3.3.3.4.4, \mathcal{O}_1) &= 2n = \{2, 4, 6, 8, 10, \dots\} \\
k_{pmg}(3.3.3.4.4, \mathcal{O}_1) &= 2n = \{2, 4, 6, 8, 10, \dots\} \\
k_{cm\bar{m}}(3.3.3.4.4, \mathcal{O}_1) &= (2n - 1), 4n = \{1, 3, 4, 5, 7, 8, 9, \dots\} \\
K(3.3.3.4.4, \mathcal{O}_1) &= k_{pm}(3.3.3.4.4, \mathcal{O}_1) \cup k_{cm}(3.3.3.4.4, \mathcal{O}_1) \cup k_{pmm}(3.3.3.4.4, \mathcal{O}_1) \cup k_{pmg}(3.3.3.4.4, \mathcal{O}_1) \\
&\quad \cup k_{cm\bar{m}}(3.3.3.4.4, \mathcal{O}_1) \\
K(3.3.3.4.4, \mathcal{O}_1) &= \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, \dots\}
\end{aligned}$$

The square orbit $\mathcal{O}_2$ possesses a higher symmetry stabilizer $\mathcal{S}_2 \cong D_2$. The compatible subgroups must strictly contain the stabilizer, limiting the valid subgroups to pmm and $cm\bar{m}$. For $\mathcal{J} \cong pmm$ (order 4), the point group index is unity ($I_p = 1$). However, the geometric mismatch between the primitive rectangular lattice of pmm and the rhombic lattice of the parent $cm\bar{m}$ imposes translation cost of $I_t = 2$. With the scaling function $f(n) = n$ for this specific lattice, the resulting indices are of the form $k_{pmm} = 2n$, generating the set $\{2, 4, 6, 8, 10, \dots\}$. For $\mathcal{J} \cong cm\bar{m}$ (order 4), both I_p and I_t are equal to 1. The scaling is restricted to odd factors or quadrupled factors, generating the set $\{1, 3, 4, 5, 7, 8, \dots\}$. The perfect coloring numbers for the orbit of square tiles are:

$$\begin{aligned}
k_{pmm}(3.3.3.4.4, \mathcal{O}_2) &= 2n = \{2, 4, 6, 8, 10, \dots\} \\
k_{cm\bar{m}}(3.3.3.4.4, \mathcal{O}_2) &= (2n - 1), 4n = \{1, 3, 4, 5, 7, \dots\} \\
K(3.3.3.4.4, \mathcal{O}_2) &= k_{pmm}(3.3.3.4.4, \mathcal{O}_2) \cup k_{cm\bar{m}}(3.3.3.4.4, \mathcal{O}_2) \\
K(3.3.3.4.4, \mathcal{O}_2) &= \{1, 2, 3, 4, 5, 6, 7, \dots\}
\end{aligned}$$

The two orbits of 3.3.3.4.4 have perfect coloring numbers $2n - 1$ and $2n$ which cover all odd and even integers. Hence, the perfect coloring numbers for the 3.3.3.4.4 tiling is the set of all positive integers, that is, there is no minimum number of colors η that forces non-perfect coloring for 3.3.3.4.4.

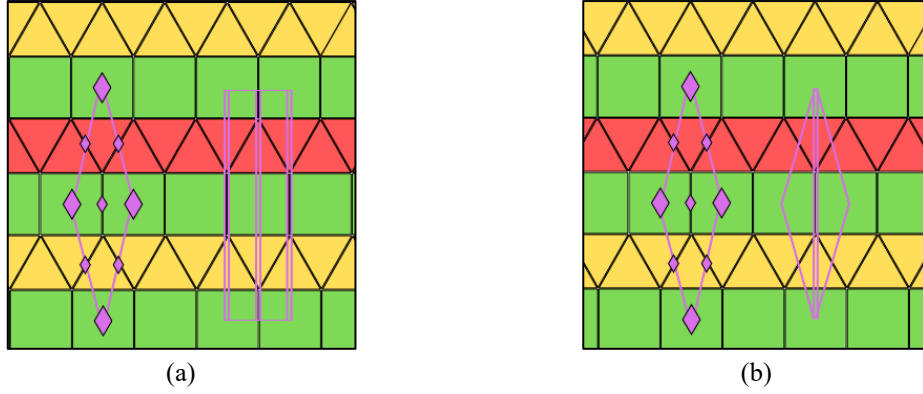


Figure 4. geometric mismatch of cmm , to pm and to cm

In Fig. 4, (a) the rhombic lattice of cmm can be fitted onto the primitive rectangular lattice of pm when factored by 2. Similarly, (b) the rhombic lattice of cmm is a natural fit onto the rhombic lattice of cm .

3.2.4. The non-regular tiling 3.3.4.3.4 ($\mathcal{G} \cong p4g$)

The 3.3.4.3.4 tiling, with symmetry group $\mathcal{G} \cong p4g$, consists of two Type II orbits, the triangular tiles ($\mathcal{O}_1$) and the square tiles ($\mathcal{O}_2$). Like the 3.3.3.4.4 tiling, the perfect coloring numbers are derived by using the index formula, however, the higher symmetry order of $p4g$, which is 8, and the square lattice constraints introduce distinct scaling behaviors.

For the triangular orbit, $\mathcal{O}_1$, the tile stabilizer is D_1 (order 2), necessitating subgroups of order 2 or higher. The analysis of compatible subgroups, pm (order 2), cm (order 2), pmm (order 4), mg (order 4), cmm (order 4), and $p4g$ (order 8), reveals different coloring varying coloring patterns. Subgroups mapping to primitive rectangular lattices incur a geometric translation cost of $I_t = 2$ and linear scaling, generating indices of the form $8n$, $4n$, and $4n$ for pm , pmm , and pmg , respectively. Conversely subgroups retaining square or rhombic symmetries (cm , cmm , $p4g$) allow for direct lattice mapping, that is, $I_t = 1$. For the rhombic lattices, the scaling functions are odd factors or quadrupled factors. For the subgroup $\mathcal{J} \cong p4g$, which is a square lattice with no reflection, the lattice scaling is quadratic and must be odd scaling preserve the 2-fold and 4-fold rotations.

$$k_{pm}(3.3.4.3.4, \mathcal{O}_1) = 8n = \{8, 16, 24, 32, 40, \dots\}$$

$$k_{cm}(3.3.4.3.4, \mathcal{O}_1) = 4(2n - 1), 4(4n) = \{4, 12, 16, 20, 28, \dots\}$$

$$k_{pmm}(3.3.4.3.4, \mathcal{O}_1) = 4n = \{4, 8, 12, 16, 20, \dots\}$$

$$k_{pmg}(3.3.4.3.4, \mathcal{O}_1) = 4n = \{4, 8, 12, 16, 20, \dots\}$$

$$k_{cmm}(3.3.4.3.4, \mathcal{O}_1) = 2(2n - 1), 2(4n) = \{2, 6, 8, 10, 14, \dots\}$$

$$k_{p4g}(3.3.4.3.4, \mathcal{O}_1) = (2n - 1)^2 = \{1, 9, 25, 49, \dots\}$$

$$K(3.3.4.3.4, \mathcal{O}_1) = k_{pm}(3.3.4.3.4, \mathcal{O}_1) \cup k_{cm}(3.3.4.3.4, \mathcal{O}_1) \cup k_{pmm}(3.3.4.3.4, \mathcal{O}_1) \cup k_{pmg}(3.3.4.3.4, \mathcal{O}_1) \\ \cup k_{cmm}(3.3.4.3.4, \mathcal{O}_1) \cup k_{p4g}(3.3.4.3.4, \mathcal{O}_1)$$

$$K(3.3.4.3.4, \mathcal{O}_1) = \{1, 2, 4, 6, 8, 9, 10, \dots\}$$

The square orbit ($\mathcal{O}_2$) is more constrained due to its stabilizer C_4 (order 4), which restricts valid subgroups to $p4$ and $p4g$ only. For $\mathcal{J} \cong p4$, the index calculation yields $I_p = 2$ and $I_t = 1$. Since the square lattice allows for quadratic scaling, and $p4$ do not have reflections, the lattice can be tilted at an angle, say if vector (a, b) , the scaling factor is $a^2 + b^2$. Hence, the valid indices are of the form $2(a^2 + b^2)$ where $a, b \geq 0$ generating the set $\{2, 4, 8, 10, \dots\}$. For $\mathcal{J} \cong p4g$, the full symmetry subgroup yields $I_p = 1$ and $I_t = 1$. The scaling is restricted to odd factors to preserve the fold rotations; hence, the valid indices are $(2n - 1)^2$ generating the set $\{1, 9, 25, 49, \dots\}$.

$$k_{p_4}(3.3.4.3.4, \mathcal{O}_2) = 2(a^2 + b^2) = \{2, 4, 8, 10, \dots\}$$

$$k_{p_{4g}}(3.3.4.3.4, \mathcal{O}_2) = (2n - 1)^2 = \{1, 9, 25, 49, \dots\}$$

$$K(3.3.4.3.4, \mathcal{O}_2) = k_{p_4}(3.3.4.3.4, \mathcal{O}_1) \cup k_{p_{4g}}(3.3.4.3.4, \mathcal{O}_1)$$

$$K(3.3.4.3.4, \mathcal{O}_1) = \{1, 2, 4, 8, 9, 10, \dots\}$$

Table 3. Sequential Verification for k of 3.3.4.3.4

Target	$k_{\mathcal{O}_1}$	$k_{\mathcal{O}_2}$
3	1	2
4	2	2
for odd $k \geq 5$	$4n$ or $2(2n - 1)$; whichever satisfies	1
for even $k \geq 6$	$4n$ or $2(2n - 1)$; whichever satisfies	2

The possible k -colorings for 3.3.4.3.4 are determined in Table 4.3. Since every integer $k \geq 3$ is valid for 3.3.4.3.4, the tiling does not have a minimum η and every positive integer k colors admit at least one perfect coloring configuration for 3.3.4.3.4.

3.2.5. The non-regular tilings 3.4.6.4 and 4.6.12 ($\mathcal{G} \cong p6m$)

The 3.4.6.4 and 4.6.12 uniform tilings exhibit a structural equivalence, sharing the parent symmetry group $\mathcal{G} \cong p6m$. As demonstrated in Fig. 5, the orbits of the 3.4.6.4 tiling (triangles, squares, and hexagons) map directly to the corresponding orbits of the 4.6.12 tiling (hexagons, squares, and dodecagons, respectively) without altering the underlying symmetry lattice. Due to its equivalence, the derivation of perfect coloring numbers for 3.4.6.4 applies generally to 4.6.12. Without loss of generality, 3.4.6.4 tiling is considered in this derivation.

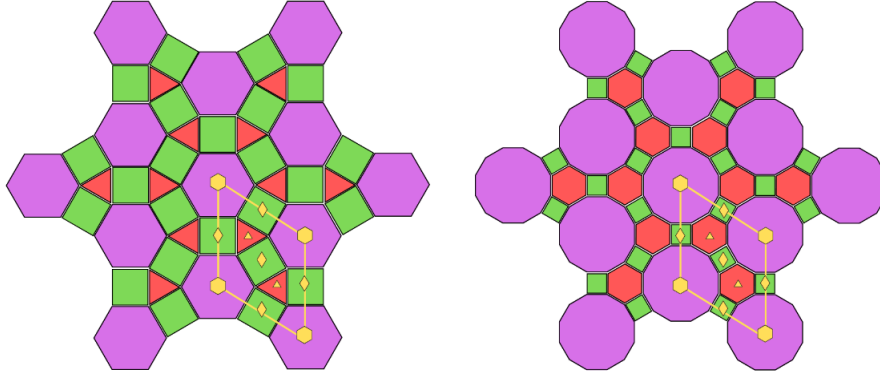


Figure 5. structural isomorphism between 3.4.6.4 and 4.6.12: mapping of rotational centers and translation unit cells

The first orbit ($\mathcal{O}_1$, triangles) and the third orbit ($\mathcal{O}_3$, hexagons) of the 3.4.6.4 tiling are classified as Type I due to their structural isomorphism with regular tilings. As illustrated in Fig.6, the triangular orbit aligns with the 3.3.3.3.3.3 lattice, while the hexagonal orbit aligns with the 6.6.6 lattice. Consequently, the perfect coloring numbers are inherited directly from the axiomatic sets established for these regular tilings. That is:

$$K(3.4.6.4, \mathcal{O}_1) = K(3.3.3.3.3.3) = \{1, 2, 4, 6, 8, 16, 18, \dots\}$$

$$K(3.4.6.4, \mathcal{O}_3) = K(6.6.6) = \{1, 3, 4, 9, 12, 16, 25, \dots\}$$

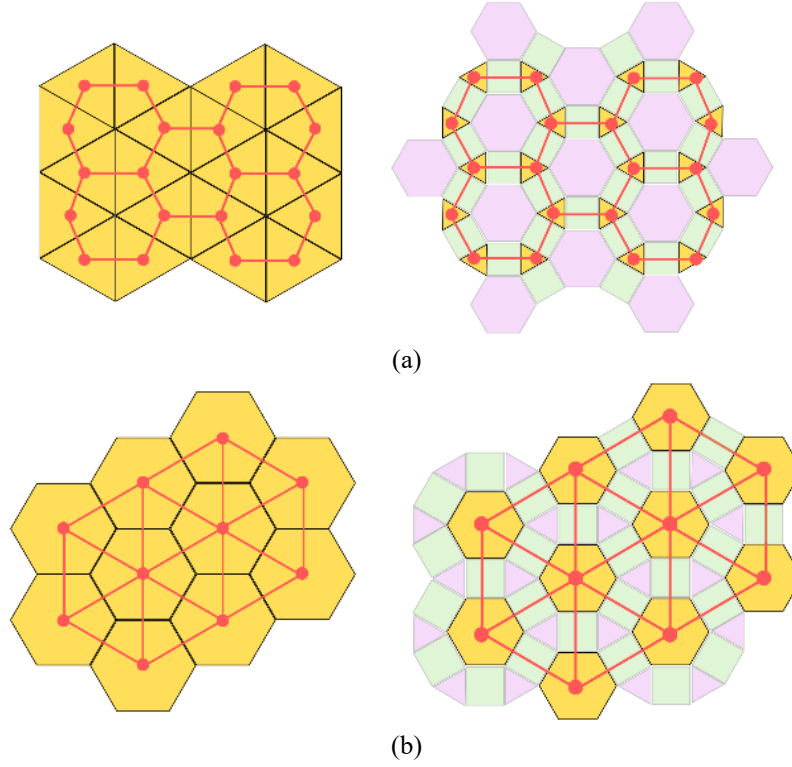


Figure 6. structural isomorphism of 3.3.3.3.3.3 and 6.6.6 to 3.4.6.4's first and third orbits

The second orbit ($\mathcal{O}_2$), consisting of square tilings, is classified as Type II. The stabilizer D_2 (order 4) requires subgroups $\mathcal{J}$ compatible with the hexagonal parent group $p6m$ (point group order 12). The valid subgroups identified are pmm , cmm , and $p6m$. For $\mathcal{J} \cong pmm$ and $\mathcal{J} \cong cmm$, both subgroups have order 4 resulting in $I_p = \frac{12}{4} = 3$. The lattices of these subgroups can fit onto the hexagonal lattice of $p6m$ with different geometric costs. Specifically, pmm requires an area scaling of $I_t = 2$, while cmm can fit naturally, $I_t = 1$. The rectangular lattice of pmm allows for linear scaling while the rhombic lattice of cmm is restricted to odd or quadrupled factors to preserve centering parity. This generates indices of the form $6n$ for pmm and $3(2n-1)$ and $3(4n)$ for cmm . The full symmetry subgroup ($p6m$) yields $I_p = 1$. However, the square orbit's centroids only align with a subset of the hexagonal lattice points, creating a specific constraint. Hexagonal lattice scaling follows a quadratic rule, and to preserve fold rotations, the factor must be odd: $(2n-1)^2$. Additionally, a rotated fitting (30°) introduces a factor of 3, yielding indices $3(2n-1)^2$. The union of these derivations provides the complete set for the square orbit.

$$k_{pmm}(3.4.6.4, \mathcal{O}_2) = 6n = \{6, 12, 18, 24, 30, \dots\}$$

$$k_{cmm}(3.4.6.4, \mathcal{O}_2) = 3(2n-1), 3(4n) = \{3, 9, 12, 15, 21, \dots\}$$

$$k_{p6m}(3.4.6.4, \mathcal{O}_2) = (2n-1)^2, 3(2n-1)^2 = \{1, 3, 9, 25, \dots\}$$

$$K(3.4.6.4, \mathcal{O}_2) = k_{pmm}(3.4.6.4, \mathcal{O}_2) \cup k_{cmm}(3.4.6.4, \mathcal{O}_2) \cup k_{p6m}(3.4.6.4, \mathcal{O}_2)$$

$$K(3.4.6.4, \mathcal{O}_2) = \{1, 3, 6, 9, 12, 15, 18, \dots\}$$

Table 4. Sequential Verification for k of 3.4.6.4

Target k	$k_{\mathcal{O}_1}$	$k_{\mathcal{O}_2}$	$k_{\mathcal{O}_3}$
4	2	1	1
5	1	3	1
for $k \geq 6; k(\bmod 3) = 0$	2	$6n$ or $3(2n - 1);$ whichever satisfies	1
for $k \geq 7; k(\bmod 3) = 1$	1	$6n$ or $3(2n - 1);$ whichever satisfies	3
for $k \geq 8; k(\bmod 3) = 2$	2	$6n$ or $3(2n - 1);$ whichever satisfies	3

From Table 4.4, every integer $k \geq 4$ is a valid perfect coloring number for 3.4.6.4. Hence, there is no minimum η for this tiling. Given the equivalence of 3.4.6.4 and 4.6.12, this conclusion also applies to the 4.6.12 tiling.

3.2.6. The non-regular tiling 3.3.3.3.6 ($\mathcal{G} \cong p6$)

The 3.3.3.3.6 tiling represents the most crystallographic complex case among the eleven uniform tilings. Governed by the plane symmetry group $\mathcal{G} \cong p6$ of order 6, this tiling, despite containing only two polygon shapes (triangles and hexagons), its tiles are partitioned into three distinct transitive orbits. The hexagons form a single orbit, denoted as $\mathcal{O}_1$ in this case. The triangles are split into two structurally distinct orbits, $\mathcal{O}_2$ consisting of triangles that share vertices with the hexagons, and $\mathcal{O}_3$ consisting of triangles that share edges with hexagons. The three orbits of the 3.3.3.3.6 tiling are all Type II orbits.

The first orbit comprising the hexagonal tiles has a tile stabilizer C_6 with order 6. This stabilizer restricts valid colorings to the parent group $p6$ itself. Since the hexagonal lattice of the coloring subgroup maps naturally to the parent lattice ($I_t = 1$) and point group index is $I_p = \frac{6}{6} = 1$, the perfect coloring numbers are determined strictly by the scaling function. For a hexagonal lattice with no reflections with basis vectors of length 1, a sublattice generated by a vector $au + bv$ has an area relative to the unit cell of $a^2 + ab + b^2$ for $a, b \geq 0$. This yields the set of Löschian numbers $\{1, 3, 4, 7, 9, 12, \dots\}$.

$$k_{p6}(3.3.3.3.6, \mathcal{O}_1) = a^2 + ab + b^2 = \{1, 3, 4, 7, 9, 12, \dots\}$$

$$K(3.3.3.3.6, \mathcal{O}_1) = \{1, 3, 4, 7, 9, 12, \dots\}$$

The second orbit consists of the “vertex” triangles, which have a stabilizer of C_3 (order 3). The compatible subgroups are $p3$ and $p6$. For $\mathcal{J} \cong p3$ (order 3), the point group index is $I_p = \frac{6}{3} = 2$ and a natural fitting of the lattices, hence, $I_t = 1$, producing indices of the form $2(a^2 + ab + b^2)$. For $\mathcal{J} \cong p6$, the point group index is $I_p = \frac{6}{6} = 1$ and fitting $p6$ onto itself is a natural fit, $I_t = 1$. Hence, the scaling function is $a^2 + ab + b^2$.

$$k_{p3}(3.3.3.3.6, \mathcal{O}_2) = 2(a^2 + ab + b^2) = \{2, 6, 8, 14, 18, \dots\}$$

$$k_{p6}(3.3.3.3.6, \mathcal{O}_2) = a^2 + ab + b^2 = \{1, 3, 4, 7, 9, 12, \dots\}$$

$$K(3.3.3.3.6, \mathcal{O}_2) = k_{p3}(3.3.3.3.6, \mathcal{O}_2) \cup k_{p6}(3.3.3.3.6, \mathcal{O}_2)$$

$$K(3.3.3.3.6, \mathcal{O}_2) = \{1, 2, 3, 4, 6, 7, 8, \dots\}$$

The third orbit consisting of “edge” triangles have a trivial stabilizer C_1 of order 1. This lack of symmetry allows compatibility with all subgroups of $p6$: $\{p1, p2, p3, p6\}$. The low-symmetry subgroups ($p1, p2, p3$) generate dense sets of indices (multiples of 6, 3, and 2, respectively) due to high rotational indices. When combined with Löschian numbers from $\mathcal{J} \cong p6$, the resulting set is extremely dense, containing $\{1, 2, 3, 4, 6, 7, 8, 9, \dots\}$ essentially all integers except specific non-Löschian primes (e.g., 5, 11).

$$k_{p1}(3.3.3.3.6, \mathcal{O}_3) = 6(a^2 + ab + b^2) = \{6, 18, 24, 42, \dots\}$$

$$k_{p2}(3.3.3.3.6, \mathcal{O}_3) = 3(a^2 + ab + b^2) = \{3, 9, 12, 21, \dots\}$$

$$k_{p3}(3.3.3.3.6, \mathcal{O}_3) = 2(a^2 + ab + b^2) = \{2, 6, 8, 14, \dots\}$$

$$k_{p6}(3.3.3.3.6, \mathcal{O}_3) = a^2 + ab + b^2 = \{1, 3, 4, 7, \dots\}$$

$$K(3.3.3.3.6, \mathcal{O}_3) = k_{p1}(3.3.3.3.6, \mathcal{O}_3) \cup k_{p2}(3.3.3.3.6, \mathcal{O}_3) \cup k_{p3}(3.3.3.3.6, \mathcal{O}_3) \cup k_{p6}(3.3.3.3.6, \mathcal{O}_3)$$

$$K(3.3.3.3.6, \mathcal{O}_3) = \{1, 2, 3, 4, 6, 7, 8, 9, \dots\}$$

Table 5. Sequential Verification for k of 3.3.3.3.6 From $k = 4$ to $k = 11$

Target k	$k_{\mathcal{T}_1}$	$k_{\mathcal{T}_2}$	$k_{\mathcal{T}_3}$
4	1	2	1
5	1	1	3
6	1	1	4
7	1	2	4
8	1	1	6
9	1	2	6
10	3	1	6
11	1	1	9
$\vdots$	$\vdots$	$\vdots$	$\vdots$

Given the arithmetic density of the sets particularly $K(3.3.3.3.6, \mathcal{O}_3)$, which contains almost all integers, and the additive flexibility provided by the other two orbits, analytical determination of the sumset is non-trivial. Therefore, the result is established through sequential computational verification. Every integer $k \geq 4$ can be successfully constructed by summing valid indices from the three orbits. Based on this verification, it is conjectured that the set of K contains all natural numbers $k \geq 4$. Consequently, there is empirical evidence that no minimum number of colors forces a non-perfect coloring for the 3.3.3.3.6 tiling.

Conjecture 4.1. The 3.3.3.3.6 tiling admits perfect colorings for all integers $k \geq 4$. Hence, no η exists in this tiling.

4. Conclusions and Recommendation

This study determined, for each of the eleven Euclidean uniform tilings, whether there exists a minimum number of colors η that forces a non-perfect coloring. The results show that the existence of such a threshold depends on the subgroup structure of the symmetry group and the interaction of orbit-level admissible indices. For six tilings, 3.3.3.3.3.3, 4.4.4.4.4, 6.6.6, 4.8.8, 3.6.3.6, 3.12.12, the set of perfect coloring numbers contains gaps, and the smallest missing integer η guarantees non-perfect coloring. For the remaining five tilings, the additive combinations of admissible orbit indices generate all integers $k \geq 3$, implying that no such minimum non-perfect threshold exists.

The analysis demonstrates that perfect colorability is governed primarily by the stabilizer subgroup and lattice structure rather than by the geometric shape of the tiles. When an orbit shares both the symmetry group and stabilizer of a regular tiling, its coloring behavior follows directly by permutation isomorphism. For non-regular cases, the derived index formula unifies point group quotients and translation lattice indices, providing a systematic method for computing admissible coloring numbers. In particular, the case of the tiling 3.3.3.3.6 reduces to an additive number-theoretic problem involving constrained quadratic forms, where empirical evidence indicates that sufficiently large integers are representable, leading to universal perfect colorability for $k \geq 3$.

Overall, the study establishes a structural criterion for determining when non-perfect coloring becomes unavoidable and clarifies the algebraic mechanisms that govern symmetry-preserving colorings in Euclidean uniform tilings.

Future work may extend this framework to k -uniform tilings, where vertex transitivity is no longer assumed and multiple vertex types coexist. Such an extension would test the generality of the orbit-based decomposition and may reveal more intricate interactions among stabilizers and subgroup indices.

Further investigation may also focus on classifying the explicit color permutation groups corresponding to each admissible index. While this study determines the number of colors required for perfect coloring or non-perfect forcing, identifying the precise permutation structures would provide a deeper algebraic characterization of color symmetry.

The subgroup index method developed here may likewise be applied to non-Euclidean settings, particularly uniform tilings of the hyperbolic plane. Examining whether analogous gap phenomena occur under hyperbolic symmetry groups would broaden the geometric scope of perfect coloring theory. Finally, the development of computational algorithms capable of generating admissible coloring numbers directly from symmetry parameters would strengthen the connection between theoretical group analysis and applications in crystallography and geometric design.

Acknowledgment

Grateful acknowledgement is hereby extended to the valuable people who have provided the researchers' much needed support to complete this paper:

To our research adviser, Dr. Kristan B. Liza, for his unwavering patience, technical guidance, and for sharing his expertise in group theory and crystallography, which were vital in the successful completion of this study;

To the members of the oral examination panel, Mr. Charly R. Lacorte, Mr. Arjay T. Altovar, and Ms. Arlene D. Faller, for their insightful critiques and suggestions that significantly improved the quality and mathematical rigor of this research;

And above all, to Almighty God, for the divine guidance, intellectual clarity, and endurance granted to the researchers throughout this journey.

References

- Coxeter, H. S. M. 1969. *Introduction to Geometry*. 2nd ed. New York: John Wiley & Sons.
- De Las Peñas, Ma. Louise Antonette N., René P. Felix, Benedict R. Gozo Jr., and G. R. Laigo. 2011. "Semi Perfect Colourings of Hyperbolic Tilings." *Philosophical Magazine* 91, no. 19–21: 2700–2708.
- Felix, René P., and L. A. Eclarin. 2014. "Transitive Perfect Colorings of 2-Uniform Tilings." *Science Diliman* 26, no. 2: 1–20.
- Grünbaum, Branko, and G. C. Shephard. 1977. "Perfect Colorings of Transitive Tilings and Patterns in the Plane." *Discrete Mathematics* 20: 235–247.
- Grünbaum, Branko, and G. C. Shephard. 1987. *Tilings and Patterns*. New York: Courier Dover Publications.
- Kaplan, Craig S. 2022. *Introductory Tiling Theory for Computer Graphics*. Boca Raton: CRC Press.

- Kim, Matthew N. 2024. "2D Crystallographic Symmetry." Matthew N. Kim (blog). November 12, 2024. <https://www.matthewnkim.com/2024/11/12/2d-crystallographic-symmetry-1/>.
- Liu, Yanxi, and Robert T. Collins. 1998. *Frieze and Wallpaper Symmetry Groups Classification Under Affine and Perspective Distortion*. Pittsburgh: Carnegie Mellon University, The Robotics Institute.
- Lockwood, Edward H., and Robert H. Macmillan. 1978. *Geometric Symmetry*. Cambridge: Cambridge University Press.
- Loquias, Manuel Joseph C., and R. B. Santos. 2023. "Perfect Precise Colorings of Plane Semiregular Tilings." *Acta Crystallographica Section A: Foundations and Advances* 79, no. 5: 440–451.
- Max, Nelson. 2023. "Constructing and Visualizing Uniform Tilings." *Computers* 12, no. 10: 208. <https://doi.org/10.3390/computers12100208>.
- Senechal, Marjorie. 1990. *Crystalline Symmetries: An Informal Mathematical Introduction*. Bristol: Springer.
- Santos, R. B., and René P. Felix. 2006. "On Perfect Colorings of Archimedean Tilings with Two Color Orbits." *Matimyas Matematika* 29, no. 3: 25–36.
- Santos, R. B., L. D. Valdez, and M. L. B. Walo. 2023. "Symmetric Perfect and Symmetric Semiperfect Colorings of Groups." *Symmetry* 15, no. 7: 1460.
- Schattschneider, Doris. 1978. "The Plane Symmetry Groups: Their Recognition and Notation." *The American Mathematical Monthly* 85, no. 6: 439–450.